

Innovation Management

Group 9 : DBC Paper s.r.l.

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Chat-GPT has been utilized to enhance the
clarity of the text



About the company...

DBC Paper, founded in 2004 in Caraglio and later relocated to Candiolo in 2010, is a leading **paper preparation** center in Italy. Known for its **state-of-the-art facilities** and **extensive industry expertise**, the company specializes in cutting and distributing paper and cardboard.

- **Net Worth:** Over \$10 million
- **Team:** A dynamic group of 10–20 dedicated professionals.



Core values...

DBC Paper's company policy is built on the " four foundational " **pillars**:

- **Search:** Focus on high-value, innovative products, continuous technological upgrades, and rigorous quality control.
- **Service:** Flexible customer service with optimized formats and robust stock of raw materials.
- **Strategy:** Diversified product portfolio and long-term partnerships with suppliers and customers.
- **Sustainability:** Commitment to ESG goals, use of FSC/PEFC-certified products, renewable energy, and waste reuse initiatives.



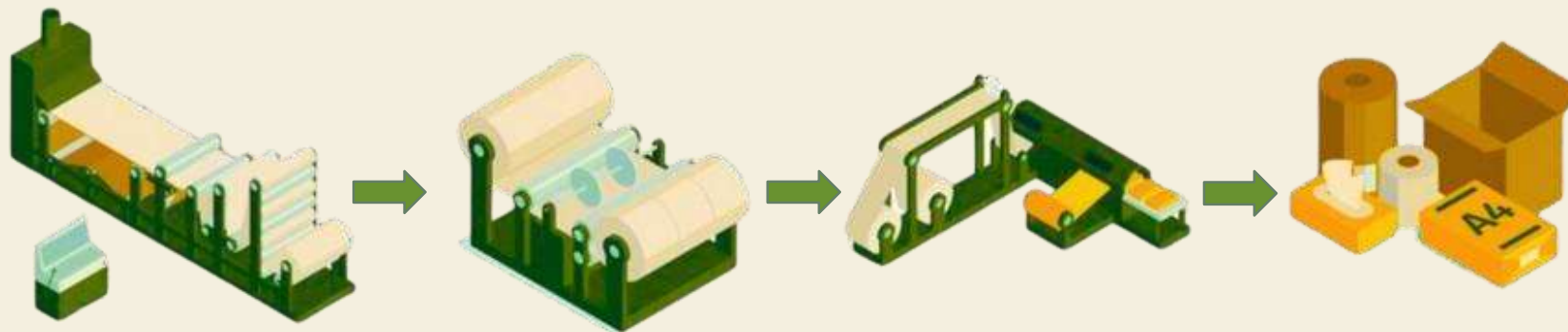
SWOT Analysis...

- **Strengths:** Strategic location, network in a niche market, certifications for sustainable management, flexibility, and medium-sized operations.
- **Weaknesses:** Complete digitalization of the process, need for an optimal balance between purchases and sales.
- **Opportunities:** Decline of offset production in favour of digital printing, certification requirements that may reduce competition.
- **Threats:** High competition in a niche market



About the production process...

DBC Paper purchases and stores various types and sizes of paper and cardboard rolls, cutting them into custom formats to meet client needs for quantity and speed. Their large warehouse capacity and strategically located production facility ensure efficiency in production and shipping. The company primarily serves printing houses and publishers, with additional activity in the light packaging sector.



Printing techniques...

- **Offset Printing**

Offset printing is an indirect method ideal for large-scale production, commonly used for newspapers, magazines, books, and promotional materials.

Printing Process: Aluminium plates transfer ink onto a rubber-coated cylinder (blanket), which then applies it to the paper substrate.

- **Digital Printing**

Digital printing is a direct process that transfers data from a digital file straight to the printer, eliminating the need for physical plates.

Printing Process: Employs laser or inkjet technology to print directly onto the material.

DBC Paper offers products suitable for both types, enabling it to operate comprehensively across the market.

Industry description...



The paper industry, one of the oldest and most prevalent manufacturing sectors, plays a **critical role in printing, packaging, and hygiene**. Despite its importance, the sector faces **significant environmental challenges**, including:

- Unregulated deforestation, particularly in regions with insufficient regulations.
- High water and chemical usage during processing.
- CO2 emissions from the production and transportation of raw and semi-finished materials.

Certifications...

The company holds two of the most important **certifications**:

- **FSC (Forest Stewardship Council):** Ensures proper forest management and product traceability.
- **PEFC (Programme for the Endorsement of Forest Certification):** Ensures sustainable forest management from ecological, economic, and social perspectives



Smart strategy board

Purpose panel

Purpose : To provide fast and customizables solutions for the distribution and cutting of paper and carboard products in Italy.

Ambition : To become the most technologically advanced cutting and distributing paper center in Italy.

Customer panel

Target Market : Small and medium printers, publishers, and carboards manufacturers.

Value Proposition : Customizable products adapted to every need and fast deliveries thanks to their location which is necessary for companies without storage

Operations panel

Partners: International high-quality raw material suppliers such as Lecta, Navigator, and Holmen.

Core competencies : Researching new high-value-products, upgrading technology, and maintaining rigorous quality control and stock management.

Finance panel

Finance objectives :

Diversification of the product portfolio, maintaining the long-term partnerships with both suppliers and clients.

Investing more in innovative solutions and eco-friendly products.

Competition and Risk Panel

Competition factors and risks:

-The rising of a paperless society reducing clientele

-The arrival of new legislations concerning deforesting increasing the price of supplies.

-The biggest paper distributor in Italy controls the majority of the market meaning they can cut their prices and steal customers.

Resource panel

IT Systems and data: Sensors use for efficient process and stock management.

Infrastructure: Cutting-edge machines placed in 2 moderns facilities

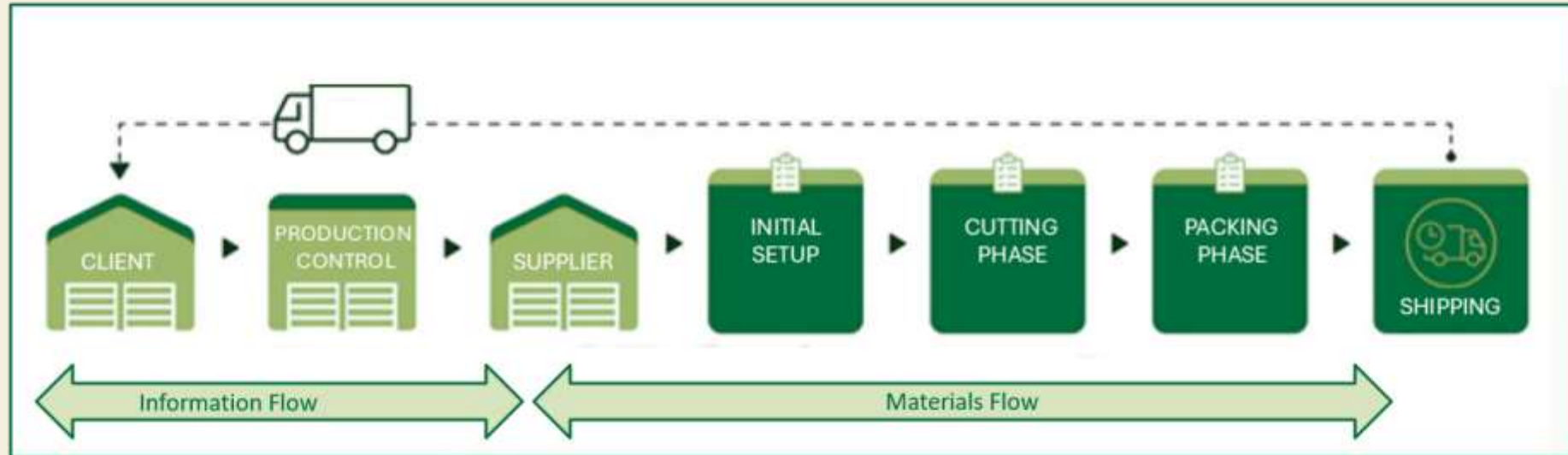
People & Talent : Small team of around 15 skilled and motivated employees.

Culture, values leadership : Very close relation with their clients.

AS-IS representation: Business Model Canvas



Value Stream Map



- **Order Processing:** Handled via phone by production control.
- **Storage:** Organized in the Initial Setup, with a six-month average holding time.
- **Quality Control:** Conducted during cutting and packaging phase.
- **Information flow:** takes 0 - 4 hours
- **Materials flow:** takes 1 day - 2 weeks
- **Shipping:** takes 1-2 days

How could we improve the business with data?

Current Role of Data:

Order Processing is managed manually via phone by production control, and also usage of digital tools for tracking or analytics.

For **Inventory Management**, data is used for basic stock tracking with a six-month average holding time. Also there is lack of predictive analytics or real-time updates.



Proposed Role of Data:

We propose **Automate Order Processing** using CRM or ERP systems to streamline workflows and enhance accuracy, and also using real-time dashboards for tracking production and inventory.

The company can use **Predictive Analytics** in such a way that uses historical data to forecast demand, optimize stock levels, and reduce holding time. Also, using data would be useful for predictive maintenance of machinery to minimize downtime.

Objective of the system...

Anticipating Client Needs

- Prediction of future clients orders by advanced time series forecasting models
- Proactive service



Optimizing Operations

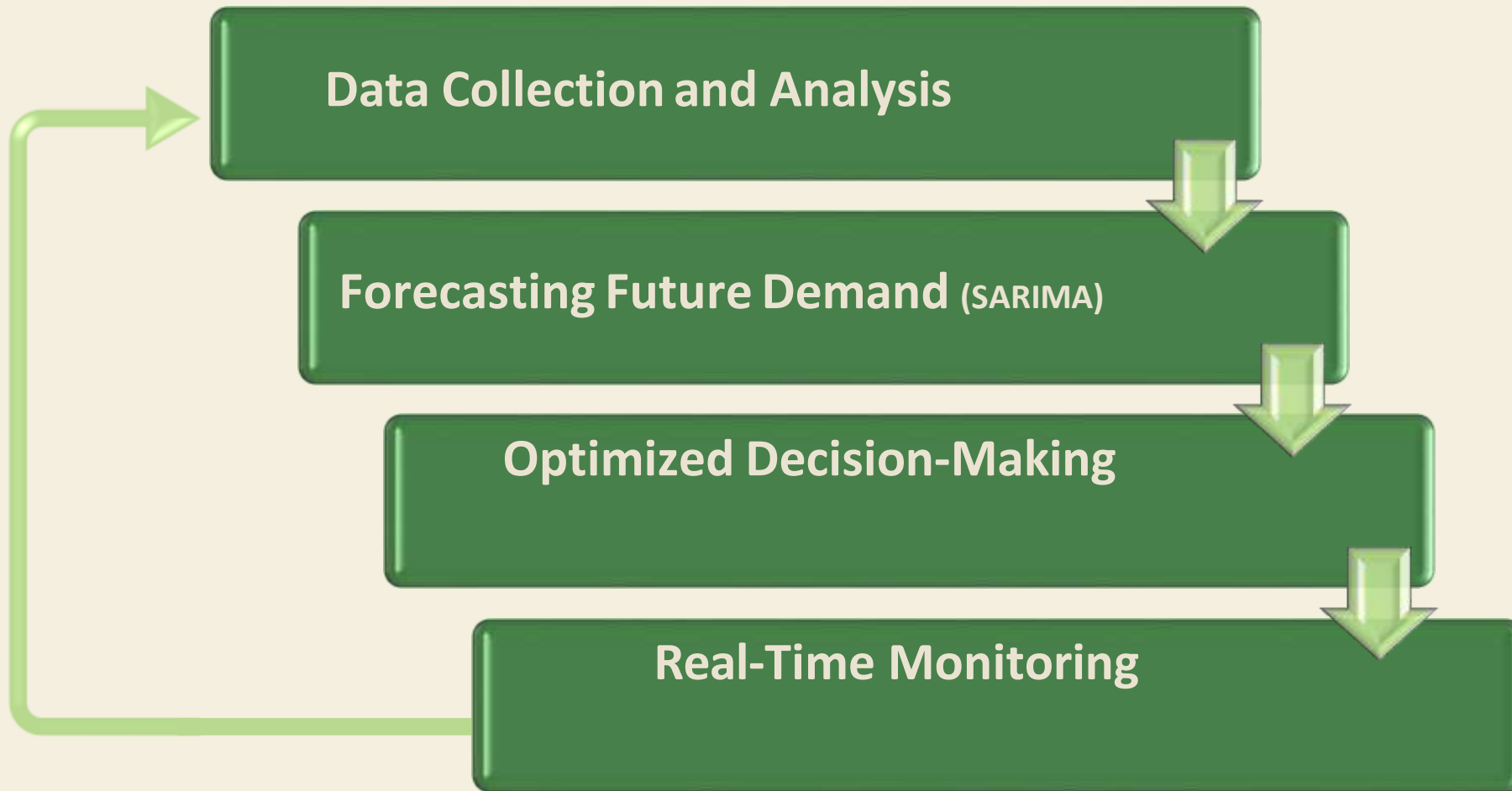
- Minimize paper waste
- Improve workflow efficiency by aligning production with data-driven insights.



Enhancing Reputation

- Build trust and reliability by:
- delivering fast
 - Tailored solutions
 - Cost efficiency

How the system works?



Key features of the system...

Demand feature module

Predicts future client orders incorporating seasonal trends.

Inventory optimization

Recommends the best paper rolls to minimize waste and management costs according to a global view.

Client segmentation

Groups clients based on purchasing behaviour, enabling tailored services.

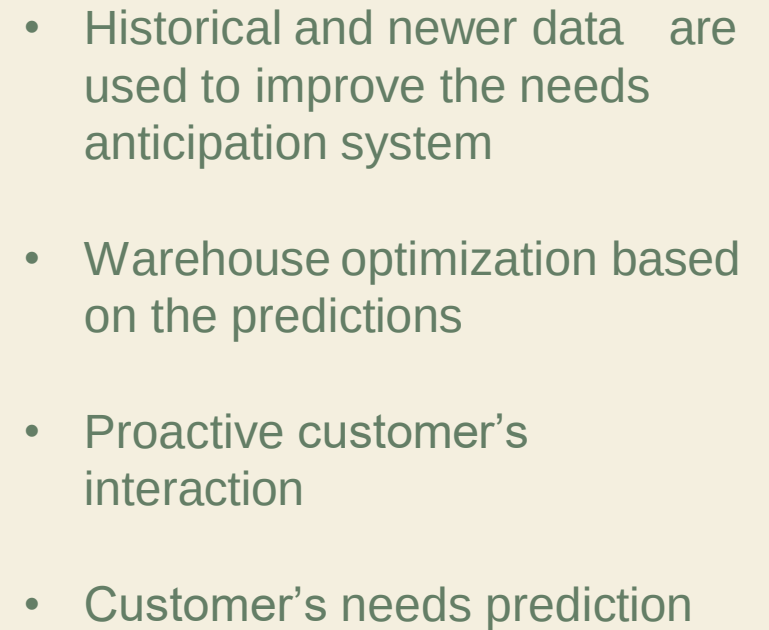
Workflow scheduling

Aligns production schedules with forecasted demand to reduce lead times.

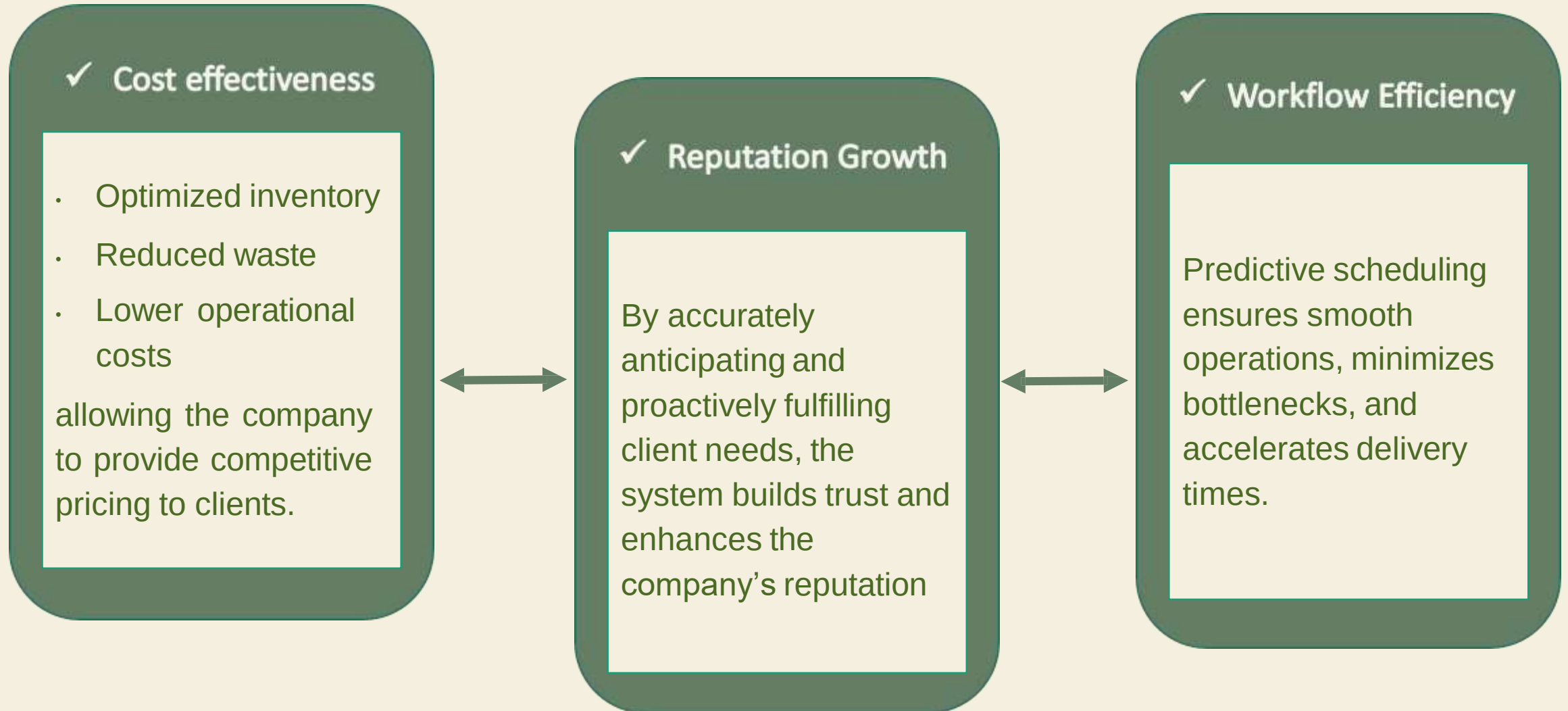
Performance dashboard

Offers real-time visualizations of key metrics

paper
dbc



How this new system provides value?



How customer care is enhanced?

- **Proactive Engagement:** Clients are notified about anticipated needs or order suggestions based on forecasts, demonstrating a deep understanding of their requirements.
- **Tailored Solutions:** Segmentation and behavior analysis allow for customized services and promotions, enhancing client satisfaction.
- **Improved Reliability:** With accurate forecasting and streamlined operations, clients can trust the company to deliver on time, every time.



How This System Enhances Sustainability and Reduces Waste?

Optimized Material Usage:

By selecting the ideal paper rolls for forecasted orders, the system minimizes off-cuts and leftover materials.

Waste Analytics:

Provides insights into waste patterns, enabling the company to implement recycling or upcycling solutions effectively.

Energy Efficiency:

Aligning production schedules with demand reduces energy consumption from unnecessary machine runs.

Sustainability Reporting:

Tracks and reports on waste reduction and sustainability efforts, strengthening the company's eco-friendly credentials.

Data science tools and technical requirements

Data storage and management:

- The AWS and RDS databases could be use to store orders and inventory
- APIs would connect systems like CRM or ERP with predictive analytics models.

Data analysis & visualization:

- The Python library Pandas can be used to analyze historical data and implement SARIMA models for future forecast.
- Power BI with APIs will give us real-time dashboards for easy access to the data.

Automatization and implementation:

- Machine learning framework like TensorFlow with integrated IoT sensors could be used to monitor equipment health and reduce downtime thus helping workflow automation.
- Implementing theses tools requires robust infrastructure. The AWS cloud service is absolutely necessary for storage and processing and installing IoT sensors on the machines would feed the data into the system

TO-BE representation: Business Model Canvas



Note: The brown ones are the changes made in the company.